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EXP NO: 6

Case Study on Food Delivery Applications – Positive Test Cases

Objective

To study the requirements of Food Delivery Applications and prepare positive test scenarios and test cases to verify the proper functioning of the application.

Features to be Tested

1. User Login and Authentication.
2. Restaurant Search and Selection.
3. Food Menu and Item Selection.
4. Add Food Items to Cart.
5. Apply Offers and Discounts.
6. Delivery Address Management.
7. Online Payment.
8. Order Confirmation and Tracking.
9. Order History.
10. Profile and Review Management.

Bug ID and Bug Name

Bug ID Bug Name

FD 001 Valid User Login

FD 002 Restaurant Search

FD 003 Food Menu Display

FD 004 Add Food Items to Cart

FD 005 Cart Quantity Update

FD 006 Apply Valid Coupon

FD 007 Delivery Address Selection

FD 008 Successful Payment

FD 009 Order Confirmation

FD 010 Order Tracking

FD 011 Order History

FD 012 Rating and Review

BUG REPORT

Bug Id: FD 001

Bug Description: Valid User Login

Steps to reproduce:

1. Launch the food delivery application.
2. Enter the valid username and password.
3. Choose the Login button.

Expected Result: Home screen should appear.

Actual Result: Home screen appears as expected.

Status: Pass.

Bug Id: FD 002

Bug Description: Restaurant Search

Steps to reproduce:

1. Log in to the food delivery application.
2. Provide a valid delivery location.
3. Choose Search Restaurants.
4. Select an available restaurant.

Expected Result: Available restaurants should appear based on the delivery location.

Actual Result: Restaurants appear as expected.

Status: Pass.

Bug Id: FD 003

Bug Description: Food Menu Display

Steps to reproduce:

1. Launch the food delivery application.
2. Select a restaurant.
3. Choose the Menu section.
4. View the available food items.

Expected Result: Food items, prices and descriptions should appear.

Actual Result: Food menu appears as expected.

Status: Pass.

Bug Id: FD 004

Bug Description: Add Food Items to Cart

Steps to reproduce:

1. Open a restaurant.
2. Select an available food item.
3. Choose Add to Cart.

4. Launch the cart.

Expected Result: Selected food item should be added to the cart.

Actual Result: Food item is added properly to the cart.

Status: Pass.

Bug Id: FD 005

Bug Description: Cart Quantity Update

Steps to reproduce:

1. Add food items to the cart.

2. Launch the cart.

3. Increase or decrease the quantity of selected food items.

4. Review the updated total amount.

Expected Result: Updated quantity and total amount should appear correctly.

Actual Result: Quantity and total amount are updated correctly.

Status: Pass.

Bug Id: FD 006

Bug Description: Apply Valid Coupon

Steps to reproduce:

1. Add food items to the cart.

2. Launch the coupon section.

3. Provide a valid coupon code.

4. Choose Apply.

Expected Result: Valid coupon should be applied and the discount should appear.

Actual Result: Coupon is applied properly.

Status: Pass.

Bug Id: FD 007

Bug Description: Delivery Address Selection

Steps to reproduce:

1. Launch the cart.

2. Choose Proceed to Checkout.

3. Provide a valid delivery address.

4. Choose Save Address.

Expected Result: Delivery address should be saved properly.

Actual Result: Delivery address is saved properly.

Status: Pass.

Bug Id: FD 008

Bug Description: Successful Payment

Steps to reproduce:

1. Proceed to checkout.

2. Select a valid payment method.

3. Enter valid payment details.

4. Choose Pay Now.

Expected Result: Payment should finish without errors.

Actual Result: Payment finishes properly.

Status: Pass.

Bug Id: FD 009

Bug Description: Order Confirmation

Steps to reproduce:

1. Complete the payment process.

2. Launch the order confirmation page.

3. Review the order details.

Expected Result: Order ID, ordered food items and total amount should appear.

Actual Result: Order confirmation details appear as expected.

Status: Pass.

Bug Id: FD 010

Bug Description: Order Tracking

Steps to reproduce:

1. Log in to the application.

2. Choose My Orders.

3. Select the current order.

4. Review the order tracking status.

Expected Result: Current order status should appear.

Actual Result: Order tracking details appear as expected.

Status: Pass.

Bug Id: FD 011

Bug Description: Order History

Steps to reproduce:

1. Launch the Profile section.

2. Choose Order History.

3. View the previous orders.

Expected Result: Previous order details should appear.

Actual Result: Order history appears as expected.

Status: Pass.

Bug Id: FD 012

Bug Description: Rating and Review

Steps to reproduce:

1. Open a completed order.
2. Choose Rate and Review.
3. Select a valid rating.
4. Enter a review.
5. Choose Submit.

Expected Result: Rating and review should be submitted properly.

Actual Result: Rating and review are submitted properly.

Status: Pass.

Result

Thus, the requirements of the Food Delivery Application were identified and analyzed. The positive test scenarios and test cases were prepared to verify the major functionalities of the application, including user login, restaurant search, food selection, cart management, payment, order tracking and review management.

EXP NO: 7

Case Study on Food Delivery Applications – Negative Test Cases

Objective

To study the requirements of Food Delivery Applications and prepare negative test scenarios and test cases to verify how the application handles invalid inputs, incorrect data and unsuccessful user actions.

Features to be Tested

1. Home Page and User Login.
2. Food Order and Favourite Items.
3. Restaurant Awards.
4. Zomato UPI and Zomato Gold.
5. Events and Event Tickets.
6. Dining Transactions and Rewards.
7. Money Gift Cards.
8. Language Selection and Feedback.
9. Profile Management and Reviews.
10. User Ratings.

Negative Test Scenarios

1. Home Page: Verify the application behavior when an incorrect email address or password is entered.
2. Food Order: Confirm whether incorrect food items are displayed when accessing favourite orders.

3. Restaurant Awards: Confirm whether fake or incorrect restaurant names are displayed in the restaurant awards section.
4. Zomato UPI: Confirm whether incorrect UPI transactions or insufficient balance messages are handled properly.
5. Events: Confirm whether incorrect event tickets are displayed.
6. Dining Transactions: Confirm whether money is incorrectly debited from the bank account during dining reward bookings.
7. Money Gift Cards: Confirm whether unauthorized users can access or purchase gift cards.
8. Language Selection: Confirm whether the application fails to send feedback after changing the language.
9. Profile Management: Confirm whether users are unable to add reviews or upload profile photos.
10. User Ratings: Confirm whether incorrect or missing ratings are displayed.

Test Environment

- Approach: Manual Approach
- Test Data: Username / Password
- Test Environment: Operating System / Browser
- Training Needed: Not Required
- Risk: No

BUG ID AND BUG NAME

1. FD 001 – Invalid Login Credentials
2. FD 002 – Incorrect Food Items in Favourite Orders
3. FD 003 – Fake Restaurant Names
4. FD 004 – Incorrect UPI Transaction
5. FD 005 – Invalid Event Tickets
6. FD 006 – Incorrect Dining Transaction
7. FD 007 – Unauthorized Gift Card Access
8. FD 008 – Feedback Submission Failure
9. FD 009 – Profile Review Failure
10. FD 010 – Incorrect User Rating

BUG REPORT

Bug Id: FD 001
Bug Description: Invalid Login Credentials
Steps to reproduce:
1. Launch the food delivery website.
2. Enter an incorrect email address or password.

3. Choose the Login button.

4. Review the login response.

Expected Result: Invalid email or password message should appear.

Actual Result: User should not be allowed to log in using invalid login details.

Status: Pass.

Bug Id: FD 002

Bug Description: Incorrect Food Items in Favourite Orders

Steps to reproduce:

1. Launch the food delivery application.

2. Choose Food Order.

3. Select Favourite Orders.

4. Review the displayed food items.

Expected Result: Only the food items added to favourites should appear.

Actual Result: Incorrect food items are displayed in the favourite orders section.

Status: Pass.

Bug Id: FD 003

Bug Description: Fake Restaurant Names

Steps to reproduce:

1. Launch the food delivery application.

2. Choose Restaurant Awards.

3. Select Winning Restaurants.

4. Confirm whether the restaurant names are correct.

Expected Result: Only genuine winning restaurant names should appear.

Actual Result: Fake or incorrect restaurant names are displayed.

Status: Pass.

Bug Id: FD 004

Bug Description: Incorrect UPI Transaction

Steps to reproduce:

1. Launch the food delivery application.

2. Choose Zomato UPI.

3. Review the UPI activity.

4. Verify the transaction details and wallet balance.

Expected Result: Incorrect transactions should not be accepted, and insufficient balance messages should appear when applicable.

Actual Result: Incorrect transaction acceptance or insufficient balance message may appear.

Status: Pass.

Bug Id: FD 005

Bug Description: Invalid Event Tickets

Steps to reproduce:

1. Launch the food delivery application.
2. Choose Events.
3. Select Your Events.
4. Review the event tickets displayed on the screen.

Expected Result: Only valid event tickets should appear.

Actual Result: Wrong event tickets are displayed on the screen.

Status: Pass.

Bug Id: FD 006

Bug Description: Incorrect Dining Transaction

Steps to reproduce:

1. Launch the food delivery application.
2. Choose Dining.
3. Select Dining Rewards.
4. Book a dining slot.
5. Review the bank transaction details.

Expected Result: Money should be debited only after a successful booking, and the correct dining reward should be applied.

Actual Result: Money may be debited incorrectly from the bank account without a successful booking.

Status: Pass.

Bug Id: FD 007

Bug Description: Unauthorized Gift Card Access

Steps to reproduce:

1. Launch the food delivery application.
2. Choose Money.
3. Select Buy Gift Cards.
4. Try to access the gift card section without valid authorization.

Expected Result: Unauthorized users should not be allowed to access or purchase gift cards.

Actual Result: Unauthorized access to the gift card section may occur.

Status: Pass.

Bug Id: FD 008

Bug Description: Feedback Submission Failure

Steps to reproduce:

1. Launch the application.
2. Select the language option.

3. Choose a different language.

4. Launch the feedback section.

5. Enter feedback and click Submit.

Expected Result: Feedback should be submitted properly.

Actual Result: Application is unable to send the feedback.

Status: Pass.

Bug Id: FD 009

Bug Description: Profile Review Failure

Steps to reproduce:

1. Launch the Profile section.

2. Select Add Review.

3. Enter the review details.

4. Try to upload a profile photo.

5. Choose Submit.

Expected Result: Review and profile photo should be submitted properly.

Actual Result: Application is unable to submit the review or upload the photo.

Status: Pass.

Bug Id: FD 010

Bug Description: Incorrect User Rating

Steps to reproduce:

1. Launch the food delivery application.

2. Navigate to Your Ratings.

3. Select a previously rated food item.

4. Confirm whether the submitted rating is displayed correctly.

Expected Result: User rating should appear correctly.

Actual Result: Incorrect or missing user rating is displayed.

Status: Pass.

Result

Thus, the requirements of the Food Delivery Application were identified and analyzed. Negative test scenarios and test cases were prepared to verify the application's behavior when invalid credentials, incorrect information, unauthorized access and unsuccessful transactions are encountered.

EXP NO: 8

Case Study on Banking Applications – Positive Test Cases

Objective

To study the requirements of Banking Applications and prepare positive test

scenarios and test cases to verify the proper functioning of the banking application.

Features to be Tested

1. User Registration.
2. User Login.
3. Account Management.
4. Fund Transfer.
5. Bill Payment.
6. Account Security.
7. Customer Support.
8. Mobile Banking.
9. Transaction Notifications.
10. Integration with External Services.

Positive Test Scenarios

1. User Registration: Check that a user can properly register using correct information and a unique username.
2. User Login: Check that a registered user can log in using correct login details.
3. Account Management: Check that users can create bank accounts, view account details and update personal information.
4. Fund Transfer: Check that users can properly transfer money between bank accounts.
5. Bill Payment: Check that users can pay electricity, water and telephone bills properly.
6. Account Security: Check that users can update passwords and enable two-factor authentication.
7. Customer Support: Check that users can contact customer support through chat, email and phone.
8. Mobile Banking: Check that the mobile application provides a user-friendly experience and supports banking functionalities.
9. Transaction Notifications: Check that users receive notifications for successful transactions and account changes.
10. Integration with External Services: Check that the banking application integrates properly with external payment gateways and services.

Test Environment

- Approach: Manual Approach
- Test Data: Username and Password
- Test Environment: Laptop / Mobile Phone
- Training Needed: No
- Risk: No

BUG ID AND BUG NAME

1. BA 001 – User Registration
2. BA 002 – User Login
3. BA 003 – Account Management
4. BA 004 – Fund Transfer
5. BA 005 – Bill Payment
6. BA 006 – Account Security
7. BA 007 – Customer Support
8. BA 008 – Mobile Banking
9. BA 009 – Transaction Notifications
10. BA 010 – Integration with External Services

BUG REPORT

Bug Id: BA 001

Bug Description: User Registration

Steps to reproduce:

1. Launch the banking application.
2. Choose the Registration option.
3. Provide a valid name, email address and password.
4. Enter a unique username.
5. Choose Register.

Expected Result: User should be properly registered with correct information.

Actual Result: User is registered properly.

Status: Pass.

Bug Id: BA 002

Bug Description: User Login

Steps to reproduce:

1. Launch the banking application.
2. Enter the registered email address and password.
3. Choose Login.
4. Confirm whether the user is logged in properly.

Expected Result: User should be able to log in using correct login details.

Actual Result: User is logged in properly.

Status: Pass.

Bug Id: BA 003

Bug Description: Account Management

Steps to reproduce:

1. Log in to the banking application.

2. Select the Account Management option.

3. Create a new bank account.

4. View account details, including balance and transaction history.

5. Update personal information.

Expected Result: User should be able to create an account, view account details and update information properly.

Actual Result: Account details are displayed and updated properly.

Status: Pass.

Bug Id: BA 004

Bug Description: Fund Transfer

Steps to reproduce:

1. Log in to the banking application.

2. Select the Fund Transfer option.

3. Enter the receiver's valid account number.

4. Enter the transfer amount.

5. Confirm the transaction.

Expected Result: Money should be transferred properly, and the amount should be accurately reflected in both sender and receiver accounts.

Actual Result: Fund transfer finishes properly.

Status: Pass.

Bug Id: BA 005

Bug Description: Bill Payment

Steps to reproduce:

1. Log in to the banking application.

2. Select the Bill Payment option.

3. Choose electricity, water or telephone bill.

4. Enter valid bill details.

5. Confirm the payment.

Expected Result: Bill payment should finish without errors, and the transaction should be reflected in the user's transaction history and biller's record.

Actual Result: Bill payment finishes properly.

Status: Pass.

Bug Id: BA 006

Bug Description: Account Security

Steps to reproduce:

1. Log in to the banking application.

2. Launch the Account Security section.

3. Select Change Password.

4. Enter the current password and a valid new password.

5. Confirm the new password.

6. Enable two-factor authentication.

Expected Result: New password should be updated properly and two-factor authentication should be enabled.

Actual Result: Password is updated and two-factor authentication is enabled properly.

Status: Pass.

Bug Id: BA 007

Bug Description: Customer Support

Steps to reproduce:

1. Log in to the banking application.

2. Launch the Customer Support section.

3. Select chat, email or phone support.

4. Submit a valid support request.

5. Review the response.

Expected Result: Customer support request should be submitted and an appropriate response should be received.

Actual Result: Customer support request is submitted correctly.

Status: Pass.

Bug Id: BA 008

Bug Description: Mobile Banking

Steps to reproduce:

1. Install and open the mobile banking application.

2. Log in using correct login details.

3. Navigate through the available banking features.

4. Review the application on supported mobile devices.

Expected Result: Mobile banking application should provide a user-friendly experience and all major banking functionalities should work properly.

Actual Result: Mobile banking application works properly.

Status: Pass.

Bug Id: BA 009

Bug Description: Transaction Notifications

Steps to reproduce:

1. Log in to the banking application.

2. Perform a successful fund transfer.

3. Review the transaction notification.

4. Verify the transaction details.

5. Check notification settings.

Expected Result: User should receive accurate notifications for successful transactions and account changes.

Actual Result: Transaction notifications are received properly with accurate details.

Status: Pass.

Bug Id: BA 010

Bug Description: Integration with External Services

Steps to reproduce:

1. Launch the banking application.

2. Select an external payment service.

3. Enter valid payment details.

4. Complete the payment.

5. Confirm whether the transaction information is updated.

Expected Result: Banking application should integrate smoothly with external services and exchange transaction data properly.

Actual Result: External service integration finishes properly.

Status: Pass.

Result

Thus, the requirements of the Banking Application were identified and analyzed. Positive test scenarios and test cases were prepared to verify the major functionalities, including user registration, login, account management, fund transfer, bill payment, account security, customer support and transaction notifications.

EXP NO: 9

CASE STUDY ON BANKING APPLICATIONS – NEGATIVE TEST CASES

Objective:

To study the requirements of Banking Applications and to prepare negative test scenarios and test cases to verify how the application handles invalid inputs, incorrect credentials, and unsuccessful transactions.

Features to be Tested

1. User Registration

2. User Login

3. Account Management

4. Bill Payment

5. Account Security

Negative Test Scenarios

1. Check that the application does not allow registration with invalid email addresses, weak passwords, or missing mandatory fields.
2. Check that the application displays an suitable error message when a user enters incorrect login credentials.
3. Check that the application does not allow users to access account details using invalid account information.
4. Check that the application displays an error when a bill payment fails or the bill details are invalid.
5. Check that the application rejects passwords that do not meet security requirements.

Test Environment

- Approach: Manual Approach
- Test Data: Invalid username, password, email, and account details
- Environment: Laptop and Mobile Phone
- Training Needed: NO
- Risk: NO

Bug ID and Bug Name

- BA 001 – Invalid User Registration
- BA 002 – Invalid User Login
- BA 003 – Invalid Account Management
- BA 004 – Bill Payment Failure
- BA 005 – Invalid Password Security

BUG REPORTS

Bug ID: BA 001

Bug Description: User Registration – Application should reject invalid registration details.

Steps to Reproduce:

Step 1 – Check that the application does not allow users to register with an invalid email address.

Step 2 – Check that mandatory fields cannot be left empty.

Step 3 – Check that the application rejects duplicate usernames and weak passwords.

Expected Result: The application should display suitable error messages for invalid registration details and should not create the account.

Actual Result: The application rejects invalid registration details and displays appropriate error messages.

Status: Pass

Bug ID: BA 002

Bug Description: User Login – Application should reject incorrect login credentials.

Steps to Reproduce:

Step 1 – Enter an invalid username or email address.
Step 2 – Enter an incorrect password and click the Login button.
Step 3 – Check that an suitable error message is displayed.
Step 4 – Check that the user is not allowed to access the account with invalid login details.
Expected Result: The application should reject invalid login details and display an appropriate error message.
Actual Result: The application rejects the login attempt and displays an error message indicating that the credentials are incorrect.
Status: Pass
Bug ID: BA 003
Bug Description: Account Management – Application should not allow access using invalid account information.
Steps to Reproduce:
Step 1 – Enter an invalid account number while accessing account details.
Step 2 – Check that the application does not display account details for an invalid account.
Step 3 – Try to update account information using invalid details.
Expected Result: The application should reject invalid account information and display an suitable error message.
Actual Result: The application does not display account details and rejects the invalid information.
Status: Pass
Bug ID: BA 004
Bug Description: Bill Payment – Application should handle unsuccessful bill payments correctly.
Steps to Reproduce:
Step 1 – Enter an invalid bill reference number or incorrect bill details.
Step 2 – Try to make a payment using incorrect information.
Step 3 – Check that the application displays an suitable error message.
Step 4 – Check that the failed payment is not recorded as a successful transaction.
Expected Result: The application should reject the invalid bill payment and display an suitable error message. The unsuccessful transaction should not appear as a successful payment in the transaction history.
Actual Result: The application rejects the invalid payment and does not record it as a successful transaction.
Status: Pass
Bug ID: BA 005
Bug Description: Account Security – Application should reject passwords that do not satisfy security requirements.

Steps to Reproduce:

Step 1 – Enter the current password incorrectly while attempting to change the password.

Step 2 – Enter a new password that does not meet the required length and complexity.

Step 3 – Check that the application rejects the invalid password.

Step 4 – Check that the application displays an suitable error message.

Expected Result: The application should reject incorrect current passwords and weak new passwords. It should display an suitable error message without updating the password.

Actual Result: The application rejects the incorrect password and does not allow the user to set a weak password.

Status: Pass

Result: Thus, negative test scenarios and test cases for the Banking Application were prepared to verify how the application handles invalid registration details, incorrect login credentials, invalid account information, unsuccessful bill payments, and weak passwords.

EXP NO: 10

LIBRARY MANAGEMENT SYSTEM – TEST PLAN

Objective:

To develop a comprehensive test plan for the Library Management System by identifying its major features and preparing test cases to verify the functionality of registration, book searching, borrowing, returning, book addition, report generation, security, and membership renewal.

Test Plan

Test Approach: Manual Approach

Test Environment: Web Browser

Training Needed: NO

Risk: NO

1. User Registration

Feature: User Authentication

Approach: Manual Approach

Test Data: Username and Password

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Launch the Library Management System website.

Step 2 – Provide a valid username and password.

Step 3 – Select the Register or Login button.

Step 4 – Check that the user is properly authenticated and the home page appears.

Expected Result: The user should be able to register or log in properly.

Status: Pass

2. Searching for a Book

Feature: Book Identification

Approach: Manual Approach

Test Data: Book Categories

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – On the home page, click the book category.

Step 2 – Select the required category.

Step 3 – Choose a book from the selected category.

Step 4 – Check that the selected book details are displayed.

Expected Result: The application should display the required book and its details.

Status: Pass

3. Borrowing a Book

Feature: Book Borrowing

Approach: Manual Approach

Test Data: Book Selection

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – On the home page, select the required book category.

Step 2 – Choose a book from the category.

Step 3 – Select the required book.

Step 4 – Select the Borrow option.

Step 5 – Check that the book is properly borrowed.

Expected Result: The application should allow the user to borrow an available book and update its availability status.

Status: Pass

4. Returning a Book

Feature: Book Returning

Approach: Manual Approach

Test Data: Borrowed Book Details

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Launch the borrowed books section.

Step 2 – Select the book that needs to be returned.

Step 3 – Select the Return option.

Step 4 – Check that the book is properly returned.

Expected Result: The application should update the book status and make it available for other users.

Status: Pass

5. Adding a New Book to the Library

Feature: New Book Addition

Approach: Manual Approach

Test Data: Book Availability and Details

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Select the category of the book.

Step 2 – Enter the book details and the number of copies.

Step 3 – Select the Add Book option.

Step 4 – Check that the new book is added to the library.

Expected Result: The application should properly add the new book and update the library records.

Status: Pass

6. Number of Books in the Library

Feature: Number of Books

Approach: Manual Approach

Test Data: Number of Books in Each Category

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Launch the book categories section.

Step 2 – Add or view the books available in each category.

Step 3 – Check that the total number of books is displayed.

Expected Result: The application should correctly identify and display the total number of books available in the library.

Status: Pass

7. Generating a Report of Borrowed Books

Feature: Generalized Report of Borrowing

Approach: Manual Approach

Test Data: Number of Borrowed Books and Date

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Launch the borrowed books section.

Step 2 – Select the date for which the report needs to be generated.

Step 3 – Select the Generate Report option.

Step 4 – Check that the report is generated properly.

Expected Result: The application should generate a report containing the borrowed book details for the selected date.

Status: Pass

8. Book ID Number

Feature: Identification of the Book

Approach: Manual Approach

Test Data: Book ID Number

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Select the required book.

Step 2 – Check that each book contains a unique ID number.

Step 3 – Click or search using the Book ID number.

Step 4 – Check that the correct book details are displayed.

Expected Result: The application should identify the book using its unique Book ID number.

Status: Pass

9. Security

Feature: Authentication and Access Control

Approach: Manual Approach

Test Data: User Login Credentials and Access Control

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Launch the Security option on the home page.

Step 2 – Select the required security options.

Step 3 – Configure the security settings according to the user's requirements.

Step 4 – Check that unauthorized users cannot access restricted information.

Expected Result: The application should maintain user security and protect library information from unauthorized access.

Status: Pass

10. Renewal Payment / Membership

Feature: Membership Renewal and Payment

Approach: Manual Approach

Test Data: Username and Password

Test Environment: Browser

Training Needed: NO

Risk: NO

Test Cases:

Step 1 – Select the Payment option on the home page.

Step 2 – If membership renewal is required, select the renewal option.

Step 3 – Enter the required membership details.

Step 4 – Complete the payment process.

Step 5 – Check that the membership is renewed properly.

Expected Result: The application should process the payment properly and update the membership validity.

Status: Pass

Result:

Thus, a test plan for the Library Management System was prepared by identifying its important features and defining the testing approach, test data, test environment, and test cases to verify the functionality of the application.

EXP NO: 11

CASE STUDY ON RAILWAY RESERVATION SYSTEM – POSITIVE TEST CASES

Objective:

To study the requirements of a Railway Reservation System and prepare positive test scenarios and test cases to verify the successful functioning of its major features.

Test Plan

Test Approach: Manual Approach

Test Environment: Laptop and Mobile Phone

Training Needed: NO

Risk: NO

1. User Registration

Feature: Name, Email, Mobile Number

Approach: Manual

Test Data: Valid Information, 10-digit Number

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Check that the user can properly register by entering all valid mandatory fields.

Step 2 – Check that the system accepts a valid email address and mobile number.

Expected Result: The user should be properly registered with all correct information.

Status: Pass

2. User Login

Feature: Username, Password

Approach: Manual

Test Data: Valid Credentials

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Check that a registered user can log in to the railway reservation portal properly.

Step 2 – Check that the home page is displayed after successful login.

Expected Result: The application should authenticate the registered user and allow access to the portal.

Status: Pass

3. Train Search

Feature: Source, Destination

Approach: Manual

Test Data: Chennai, Bangalore

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Enter the source and destination stations.

Step 2 – Select the Search Trains option.

Step 3 – Check that the system displays available trains between the two valid stations.

Expected Result: The application should display the available trains between the selected source and destination.

Status: Pass

4. Date Selection

Feature: Journey Date

Approach: Manual

Test Data: Future Date

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Launch the journey date selection option.

Step 2 – Select a valid future date from the calendar.

Step 3 – Check that the selected journey date is displayed correctly.

Expected Result: The application should allow the user to select a valid future journey date.

Status: Pass

5. Class Selection

Feature: SL, 3A, 2A, 1A

Approach: Manual

Test Data: 3-Tier AC

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Select the required train.

Step 2 – Choose the preferred travel class.

Step 3 – Check that seat availability is updated correctly based on the selected class.

Expected Result: The application should display the correct availability details for the selected travel class.

Status: Pass

6. Passenger Details

Feature: Name, Age, Gender

Approach: Manual

Test Data: Valid Passenger Information

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Enter the passenger's name, age, and gender.

Step 2 – Check that all passenger details are entered correctly.

Step 3 – Check that the system accepts valid passenger information.

Expected Result: The application should properly accept and save the passenger details.

Status: Pass

7. Berth Preference

Feature: Lower, Middle, Upper

Approach: Manual

Test Data: Lower Berth

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Select the preferred berth option.

Step 2 – Choose Lower, Middle, or Upper berth.

Step 3 – Check that the system records the selected berth preference.

Expected Result: The application should record and accept the passenger's berth preference.

Status: Pass

8. Payment Gateway

Feature: UPI, Net Banking

Approach: Manual

Test Data: Valid Payment Apps

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Select the payment option.

Step 2 – Choose UPI or Net Banking.

Step 3 – Check that the user is redirected to the payment gateway.

Step 4 – Complete the transaction properly.

Expected Result: The application should process the payment properly and confirm the transaction.

Status: Pass

9. PNR Generation

Feature: Digital Ticket

Approach: Manual

Test Data: Transaction ID

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Complete the ticket booking and payment process.

Step 2 – Check that a unique 10-digit PNR is generated after successful payment.

Step 3 – Check that the PNR is displayed with the booking details.

Expected Result: The application should generate a unique PNR after successful ticket booking.

Status: Pass

10. Ticket Download

Feature: PDF, SMS

Approach: Manual

Test Data: Confirmed PNR

Test Environment: Laptop, Phone

Training Needed: NO

Risk: NO

Test Case Scenarios:

Step 1 – Launch the confirmed ticket details.

Step 2 – Select the Download Ticket option.

Step 3 – Check that the user can download the ticket in PDF format.

Step 4 – Check that a confirmation SMS is received.

Expected Result: The application should allow the user to download the ticket in PDF format and receive a confirmation SMS.

Status: Pass

Result:

Thus, positive test scenarios and test cases for the Railway Reservation System were prepared to verify the successful functioning of registration, login, train search, date selection, class selection, passenger details, berth preference, payment gateway, PNR generation, and ticket download.

EXP NO: 12

CASE STUDY ON RAILWAY RESERVATION SYSTEM – NEGATIVE TEST CASES

Objective:

To study the requirements of a Railway Reservation System and prepare negative test scenarios and test cases to verify how the application handles invalid inputs, incorrect details, and unsuccessful booking operations.

The test plan below follows the same columns and test-case structure shown in your lab manual.

Test Plan

Test Approach: Manual Approach

Test Environment: Laptop and Mobile Phone

Training Needed: NO

Risk: NO

1. User Registration
   - Feature: Email Validation
   - Approach: Manual
   - Test Data: Invalid Email (without @ symbol)
   - Test Environment: Laptop, Phone
   - Training Needed: NO
   - Risk: NO

Test Case Scenarios:

Step 1 – Enter an invalid email address without the @ symbol during registration.

Step 2 – Select the Register button.

Step 3 – Check that the system displays an suitable error message.

Expected Result: The application should reject the invalid email address and display a validation error.

Status: Pass

2. User Login
   - Feature: Security
   - Approach: Manual
   - Test Data: Wrong Password
   - Test Environment: Laptop, Phone
   - Training Needed: NO
   - Risk: NO

Test Case Scenarios:

Step 1 – Enter a registered username with an incorrect password.

Step 2 – Select the Login button.

Step 3 – Check that the system denies access and displays an "Invalid Credentials" error message.

Expected Result: The application should not allow the user to log in with an incorrect password.

Status: Pass

3. Train Search

- Feature: Route Logic
- Approach: Manual
- Test Data: Same Station as Source and Destination
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter the same station as both source and destination.

Step 2 – Select the Search Trains option.

Step 3 – Check that the system displays an error message indicating that the source and destination stations are the same.

Expected Result: The application should not allow train searches when the source and destination stations are identical.

Status: Pass

4. Journey Date

- Feature: Date Validation
- Approach: Manual
- Test Data: Past Date
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Launch the journey date selection option.

Step 2 – Try to select a date that has already passed.

Step 3 – Check that the system restricts the user from selecting the past date.

Expected Result: The application should not allow users to select a past date for a new journey booking.

Status: Pass

5. Passenger Information

- Feature: Age Constraint
- Approach: Manual
- Test Data: Negative Age (-10)
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Enter passenger details during the booking process.

Step 2 – Enter a negative age such as -10.

Step 3 – Check that the system rejects the invalid age and displays an appropriate validation message.

Expected Result: The application should not accept negative or invalid passenger ages.

Status: Pass

6. Passenger Name Field

- Feature: Character Limit
- Approach: Manual
- Test Data: Numbers and Special Symbols
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Launch the passenger details section.

Step 2 – Enter numbers or special symbols in the passenger name field.

Step 3 – Check that the system rejects invalid characters in the name field.

Expected Result: The application should accept valid passenger names and reject inappropriate numbers or special symbols.

Status: Pass

7. Seat Booking

- Feature: Booking Limit
- Approach: Manual
- Test Data: 10 Passengers
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the ticket booking process.

Step 2 – Try to add more passengers than the permitted booking limit.

Step 3 – Check that the system does not allow booking beyond the maximum passenger limit.

Expected Result: The application should restrict the user from adding more passengers than the permitted limit per ticket.

Status: Pass

8. Payment Gateway

- Feature: Fund Check

- Approach: Manual
- Test Data: Expired Card
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Proceed to the payment gateway.

Step 2 – Enter expired card details.

Step 3 – Check that the system handles the payment failure gracefully.

Step 4 – Check that a "Transaction Failed" notification is displayed.

Expected Result: The application should reject the expired card and display a transaction failure message without confirming the booking.

Status: Pass

9. PNR Inquiry

- Feature: PNR Length
- Approach: Manual
- Test Data: 5-Digit PNR
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Launch the PNR enquiry section.

Step 2 – Enter a PNR number containing only 5 digits.

Step 3 – Check that the system displays an "Invalid PNR" message when the entered number does not meet the required length.

Expected Result: The application should reject an invalid or incomplete PNR number and display an suitable error message.

Status: Pass

10. Ticket Cancellation

- Feature: Cancellation Constraint
- Approach: Manual
- Test Data: Departed Train
- Test Environment: Laptop, Phone
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Launch the ticket cancellation section.

Step 2 – Select a ticket for a train that has already departed.

Step 3 – Check that the cancellation option is disabled or an suitable error message is displayed.

Expected Result: The application should not allow cancellation of a ticket after the train has departed.

Status: Pass

Result:

Thus, negative test scenarios and test cases for the Railway Reservation System were prepared to verify the application's handling of invalid email addresses, incorrect passwords, invalid routes, past journey dates, incorrect passenger details, booking limits, failed payments, invalid PNR numbers, and cancellation restrictions.

EXP NO: 13

CASE STUDY ON ONLINE EXAMINATION SYSTEM – POSITIVE TEST CASES

Objective:

To study the requirements of an Online Examination System and prepare positive test scenarios and test cases to verify the successful functioning of its major features.

Test Plan

Test Approach: Manual Approach

Test Environment: Laptop and Browser

Training Needed: NO

Risk: NO

1. Student Registration
   - Feature: Student ID, Email
   - Approach: Manual
   - Test Data: Valid Student Details
   - Test Environment: Laptop, Browser
   - Training Needed: NO
   - Risk: NO

Test Case Scenarios:

Step 1 – Check that the student can properly register using a valid institutional email address and student ID.

Step 2 – Check that all mandatory registration details are accepted.

Expected Result: The application should properly register the student with valid institutional details.

Status: Pass

2. Student Login

- Feature: Authentication Credentials
- Approach: Manual
- Test Data: Valid Username and Password
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the registered student can log in using correct login details.

Step 2 – Check that the examination dashboard is displayed after successful login.

Expected Result: The application should authenticate the student and provide access to the examination dashboard.

Status: Pass

3. Profile Verification

- Feature: Photo/ID Upload
- Approach: Manual
- Test Data: JPEG/PNG Image
- Test Environment: Laptop, Webcam
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the system properly captures or uploads the student's photograph for proctoring.

Step 2 – Check that the uploaded photograph is displayed correctly.

Expected Result: The application should properly capture or upload the student's photograph for identity verification.

Status: Pass

4. Exam Selection

- Feature: Subject List
- Approach: Manual
- Test Data: Assigned Subject
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the student can view the list of assigned examinations.

Step 2 – Select the required examination.

Step 3 – Check that the student can access only the examinations for which they are eligible.

Expected Result: The application should display the assigned examinations and allow the student to select an eligible examination.

Status: Pass

5. Instruction Page

- Feature: Guidelines
- Approach: Manual
- Test Data: Click "Agree"
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the student can read the examination instructions.

Step 2 – Select the "Agree" button after reading the guidelines.

Step 3 – Check that the system allows the student to proceed to the examination.

Expected Result: The application should display the examination guidelines and allow the student to start the exam after accepting them.

Status: Pass

6. Question Navigation

- Feature: Next/Previous
- Approach: Manual
- Test Data: Radio Buttons
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the student can move between questions using the Next and Previous buttons.

Step 2 – Check that the student can select an answer using the radio buttons.

Step 3 – Check that the student can navigate between questions and select options properly.

Expected Result: The application should allow students to navigate between questions and select their answers properly.

Status: Pass

7. Save and Mark

- Feature: Review Later
- Approach: Manual

- Test Data: Flag Feature
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the student can mark a question for review.

Step 2 – Check that the marked question appears highlighted in the question palette.

Step 3 – Check that the student can move between questions and select options properly.

Expected Result: The application should allow students to mark questions for review and identify them easily using the question palette.

Status: Pass

8. Timer Function

- Feature: Countdown
- Approach: Manual
- Test Data: System Clock
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the examination timer starts correctly when the exam begins.

Step 2 – Check that the remaining time is displayed accurately.

Step 3 – Check that the timer continues counting down during the examination.

Expected Result: The application should display the correct remaining examination time and update the countdown accurately.

Status: Pass

9. Auto-Submit

- Feature: Session End
- Approach: Manual
- Test Data: Time Expiry
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the examination is automatically submitted when the allotted time is completed.

Step 2 – Check that the student's saved answers are submitted properly.

Expected Result: The application should automatically submit the examination when the time duration is completed.

Status: Pass

10. Result Generation

- Feature: Scorecard
- Approach: Manual
- Test Data: Valid Answers
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Check that the student can view the result after the evaluation is processed.

Step 2 – Check that the scorecard displays the correct marks.

Step 3 – Check that the student can download the scorecard if the option is available.

Expected Result: The application should generate and display the student's scorecard accurately after evaluation.

Status: Pass

Result:

Thus, positive test scenarios and test cases for the Online Examination System were prepared to verify the successful functioning of student registration, login, profile verification, exam selection, instructions, question navigation, save and mark, timer function, auto-submit, and result generation.

EXP NO: 14

CASE STUDY ON ONLINE EXAMINATION SYSTEM – NEGATIVE TEST CASES

Objective:

To study the requirements of an Online Examination System and prepare negative test scenarios and test cases to verify how the application handles invalid inputs, unauthorized access, session errors, and unsuccessful examination operations.

The following test cases follow the 10 test items shown in your lab manual.

Test Plan

Test Approach: Manual Approach

Test Environment: Laptop and Browser

Training Needed: NO

Risk: NO

1. User Login

- Feature: Authentication

- Approach: Manual
- Test Data: Invalid Password
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Provide a valid username with an incorrect password.

Step 2 – Select the Login button.

Step 3 – Check that the system denies access and displays an "Incorrect Password" error after 3 failed attempts.

Expected Result: The application should reject incorrect login credentials and prevent unauthorized access.

Status: Pass

2. Exam Access

- Feature: Eligibility
- Approach: Manual
- Test Data: Expired Exam Link
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Try to access the examination using an expired exam link.

Step 2 – Check that the student cannot start the examination after the scheduled time window has closed.

Expected Result: The application should prevent students from accessing examinations after the permitted time window.

Status: Pass

3. Proctoring Check

- Feature: Webcam Access
- Approach: Manual
- Test Data: Camera Disabled
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Disable the webcam or deny camera permission in the browser.

Step 2 – Try to start the examination.

Step 3 – Check that the system prevents the examination from starting if the webcam is not detected or access is blocked.

Expected Result: The application should display an appropriate warning and prevent the examination from starting when webcam access is mandatory.

Status: Pass

4. Tab Switching

- Feature: Security
- Approach: Manual
- Test Data: Alt + Tab
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the online examination.

Step 2 – Press Alt + Tab to switch to another application or browser tab.

Step 3 – Check that the system displays a warning or automatically submits the examination if the student attempts to switch tabs.

Expected Result: The application should detect tab switching and display a warning or apply the configured examination security restrictions.

Status: Pass

5. Input Validation

- Feature: Answer Validation – Field Symbols
- Approach: Manual
- Test Data: Special Symbols
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Open a question that accepts only numeric answers.

Step 2 – Enter alphabets or special symbols in the answer field.

Step 3 – Check that the system rejects invalid characters.

Expected Result: The application should allow only numeric values in numeric-only answer fields and reject alphabets or special symbols.

Status: Pass

6. Multiple Login

- Feature: Concurrent Session

- Approach: Manual
- Test Data: Same Credentials
- Test Environment: 2 Laptops
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Log in to the examination system using the same user ID on two different devices.

Step 2 – Check that the system logs out the first session if the same user ID is logged in on another device.

Expected Result: The application should prevent unauthorized simultaneous sessions according to its session management policy.

Status: Pass

7. File Upload

- Feature: Assignment Size
- Approach: Manual
- Test Data: 100 MB File
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Launch the assignment upload section.

Step 2 – Try to upload a file that exceeds the maximum permitted size, such as a 100 MB file when the limit is 5 MB.

Step 3 – Check that the system rejects the file and displays an suitable error message.

Expected Result: The application should reject files that exceed the maximum allowed upload size.

Status: Pass

8. Network Loss

- Feature: Connectivity
- Approach: Manual
- Test Data: Wi-Fi Disconnected
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the online examination.

Step 2 – Disconnect the Wi-Fi connection during the examination.

Step 3 – Check that the system displays a "No Connection" alert.

Step 4 – Check that the system handles the interruption without losing previously saved answers.

Expected Result: The application should display a network connection alert and preserve saved examination data during network interruption.

Status: Pass

9. Submit Button

- Feature: Blank Submission
- Approach: Manual
- Test Data: No Questions Selected
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Start the online examination.

Step 2 – Leave all questions unanswered.

Step 3 – Select the Submit button.

Step 4 – Check that the system displays a warning before submitting the examination without answering any questions.

Expected Result: The application should display a confirmation or warning message when the student tries to submit without answering questions.

Status: Pass

10. URL Manipulation

- Feature: Unauthorized Access
- Approach: Manual
- Test Data: Direct Result URL
- Test Environment: Laptop, Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Copy the result page URL from an authorized session.

Step 2 – Log out of the examination system.

Step 3 – Try to access the result page directly using the copied URL.

Step 4 – Check that the student cannot access the result page without proper authentication.

Expected Result: The application should prevent unauthorized users from accessing examination results through a direct URL.

Status: Pass

Result:

Thus, negative test scenarios and test cases for the Online Examination System were prepared to verify the application's handling of incorrect login credentials, expired exam links, disabled webcams, tab switching, invalid answers, multiple login sessions, oversized file uploads, network loss, blank submissions, and unauthorized URL access.

EXP NO: 15

INVENTORY MANAGEMENT SYSTEM – TEST PLAN

Objective:

To develop a comprehensive test plan for an Inventory Management System, outlining the testing scope, design techniques, and execution strategies to ensure accurate tracking of stock, sales, and orders.

Test Plan

Test Approach: Manual Approach

Test Environment: Browser / Mobile Browser / Server

Training Needed: NO

Risk: NO

1. Admin Login
   - Feature: Authentication
   - Approach: Manual
   - Test Data: Admin ID, Password
   - Test Environment: Browser
   - Training Needed: NO
   - Risk: NO

Test Case Scenarios:

Step 1 – Launch the Inventory Management System login URL.

Step 2 – Enter the valid Admin ID and password.

Step 3 – Select the Login button and check that the dashboard appears.

Expected Result: The admin should be able to log in properly and access the dashboard.

Status: Pass

2. Add New Product
   - Feature: Inventory Entry
   - Approach: Manual
   - Test Data: Product Name, SKU, Quantity
   - Test Environment: Browser
   - Training Needed: NO
   - Risk: NO

Test Case Scenarios:

Step 1 – Select the Add Product option.

Step 2 – Enter the product details, including name, SKU, quantity, and price.

Step 3 – Click Save and check that the product is listed in the inventory.

Expected Result: The application should properly add the new product and display it in the inventory list.

Status: Pass

3. Stock Update

- Feature: Stock Level
- Approach: Manual
- Test Data: Numeric Values
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Select an existing product.

Step 2 – Increase or decrease the product quantity.

Step 3 – Check that the total stock quantity is updated correctly.

Expected Result: The application should correctly update the stock level based on the quantity added or removed.

Status: Pass

4. Supplier Management

- Feature: Supplier Information
- Approach: Manual
- Test Data: Supplier Name, Contact Details
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Add a new supplier with valid details.

Step 2 – Link the supplier to a product.

Step 3 – Save the information and check that the supplier appears in the supplier list.

Expected Result: The application should properly save supplier information and link the supplier to the selected product.

Status: Pass

5. Purchase Order

- Feature: Ordering

- Approach: Manual
- Test Data: Order ID, Supplier ID
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Create a new purchase order.

Step 2 – Add the required products and quantities to the order.

Step 3 – Check that the order status changes to "Sent".

Expected Result: The application should properly create the purchase order and update its status to "Sent".

Status: Pass

6. Sales Tracking

- Feature: Transaction
- Approach: Manual
- Test Data: Sale ID, Customer Information
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Record a new sale by entering the required product and customer details.

Step 2 – Check that the stock quantity is reduced automatically after the sale.

Step 3 – Check that the sales receipt is generated.

Expected Result: The application should record the sale, automatically update the stock quantity, and generate a receipt.

Status: Pass

7. Low Stock Alerts

- Feature: Notification
- Approach: Manual
- Test Data: Threshold Value
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Set the low-stock alert threshold at 5 units.

Step 2 – Reduce the product stock quantity to 4 units.

Step 3 – Check that an alert message appears on the dashboard.

Expected Result: The application should automatically generate a low-stock notification when the product quantity falls below the specified threshold.

Status: Pass

8. Barcode Scanning

- Feature: Product ID
- Approach: Manual
- Test Data: Barcode Serial Number
- Test Environment: Mobile / Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Scan the barcode of a product.

Step 2 – Check that the system identifies the product using its barcode.

Step 3 – Check that the corresponding product details are displayed.

Expected Result: The application should identify the product correctly and display its details after scanning the barcode.

Status: Pass

9. Report Generation

- Feature: Analytics
- Approach: Manual
- Test Data: Monthly Date Range
- Test Environment: Browser
- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Select the Monthly Sales Report option.

Step 2 – Choose the required date range, such as February 2026.

Step 3 – Generate the report and export it as a PDF or Excel file.

Expected Result: The application should generate the monthly sales report correctly and allow the admin to export it in PDF or Excel format.

Status: Pass

10. Database Backup

- Feature: Data Security
- Approach: Manual
- Test Data: SQL Backup File
- Test Environment: Server

- Training Needed: NO
- Risk: NO

Test Case Scenarios:

Step 1 – Select the Initiate Backup option.

Step 2 – Download the generated database backup file.

Step 3 – Verify the backup file and check the integrity of the stored data.

Expected Result: The application should properly create and download the database backup while maintaining data integrity.

Status: Pass

Result:

Thus, a comprehensive test plan for the Inventory Management System was prepared by identifying its major features and defining the testing approach, test data, test environment, and test cases to verify admin login, product addition, stock updates, supplier management, purchase orders, sales tracking, low-stock alerts, barcode scanning, report generation, and database backup.